

Question number	Answer	Additional guidance	Mark
6(a)	An answer showing the following steps: • correct figures taken from graph (1) • correct calculation of percentage change (for fit person) (1) • correct difference given to an appropriate number of decimal places (1)	170 – 90 ACCEPT 80 $((170 - 90) \div 170) \times 100 = 47.06 \%$ 47.06 – 16 = 31.06 % Final answer 31 % ACCEPT 31.1 %	(3)

Question number	Answer	Additional guidance	Mark
6(b)	An answer showing the following steps: • value of mx (1) • value of y (1) • correctly rounded to a whole number with units (1)	$(-65 \div 240) \times 300$ = -81.25 $= (-65 \div 240 \times 300) + 190 = 108.75$ 109 bpm Accept range 107-111	(3)

Question number	Answer	Additional guidance	Mark
6(c)	An explanation that includes three of the following points: • fit person has greater lung capacity (1) • fit person (can produce) a greater cardiac output (1) • more effective gas exchange and transport of oxygen to muscles (1) • therefore more rapid reduction of oxygen debt (1)		(3)

Question number	Answer	Additional guidance	Mark
6(d)	A description that includes the following points: • cardiovascular control centre in the medulla oblongata (1) • detects change in blood pH (following exercise) (1) • as blood pH increases (more) impulse are sent via the parasympathetic nerve (1) • slowing (the rate of impulses from) the SAN (1)	Accept increase in lactate / CO₂ Accept vagus nerve	(4)

